MASTER CLASS DELIVERABLE • VIDEO #10

Modern Web Scraping in 2026: Crawl4AI + Gemini Structured JSON

Headless Stealth Chromium, LLM-Ready Markdown Extraction & Schema-Driven Structured Output

LIBRARY: CRAWL4AI (ASYNCIO ENGINE) LLM EXTRACTOR: GEMINI 2.5 FLASH / OLLAMA PUBLISHED BY: @PARKER.RENDER

1. Why BeautifulSoup is Obsolete in 2026

Traditional HTTP requests (requests/urllib) cannot execute JavaScript or render dynamic Single Page Applications (Next.js/React). **Crawl4AI** operates a stealth headless Chromium engine that bypasses Cloudflare bot checks and automatically cleans DOM trees into token-efficient Markdown for LLMs.

2. 15-Second Installation

```
pip install crawl4ai
crawl4ai-setup
python -m playwright install --with-deps chromium
```

3. Production Scraping with Structured Schema Extraction

```
import asyncio
from pydantic import BaseModel
from crawl4ai import AsyncWebCrawler
from crawl4ai.extraction_strategy import LLMExtractionStrategy

class ProductSchema(BaseModel):
    title: str
    price: str
    in_stock: bool
    rating: float

async def main():
    strategy = LLMExtractionStrategy(
        provider="google/gemini-2.5-flash",
        api_token="YOUR_GEMINI_API_KEY",
        schema=ProductSchema.schema(),
        extraction_type="schema",
        instruction="Extract all listed products with their price and availability."
    )

    async with AsyncWebCrawler(verbose=True) as crawler:
        result = await crawler.arun(
            url="https://news.ycombinator.com",
            extraction_strategy=strategy,
            bypass_cache=True
        )

        print(result.extracted_content)
```

```
asyncio.run(main())
```

4. Key Production Features

- **Stealth Mode:** Automatically spoofs webGL vendor, user-agent headers, and browser canvas fingerprints.
- **Chunking & Markdown:** Strips navbars, footer links, and cookie banners to reduce LLM prompt token costs by up to 80%.

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